

Aero-acoustic liner applications of the broadband special acoustic absorber concept

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Aircraft exterior noise reduction is a major task for manufacturer and carrier. However, optimum sound absorption over a broad frequency range at engine level is still a challenging aspect for acoustic liner. This paper presents the special acoustic absorber concept as broadband working liner with limited space needed. Experimental as well as numerical work is presented for different applications as there are hot stream liner for exhaust integration, broadband liner for bypass integration and low frequency broadband liner for integration into a shrouded fan.

Nomenclature

c	=	speed of sound
M	=	flow Mach number
p	=	acoustic pressure
λ	=	acoustic wave length
Θ	=	acoustic resistance
ρ	=	fluid density
σ	=	porosity of facing sheet
SAA	=	special acoustic absorber
SDOF	=	single degree of freedom
DDOF	=	double degree of freedom
MDOF	=	multiple degree of freedom

I. Introduction

IN order to achieve the ACARE 2020 goals further reduction of engine noise components is still important. Due to very high bypass ratios of modern turbofan engines the relevance of core noise components as combustion, turbine and compressor raised. As a consequence of this, noise reduction measures must be effective over an increased frequency range. Usual single degree of freedom (SDOF) liners are not able to fulfil this requirement. Multiple degrees of freedom (MDOF) liners offer a higher absorption bandwidth and are capable to cover the necessary source spectrum. Special acoustic absorber (SAA) design offers the opportunity for MDOF liner integration into the engine.

II. SAA liner concept

The Special Acoustic Absorber, (SAA), principle combines a low frequency tuned Helmholtz resonator element and a high frequency tuned quarter-wave resonator part in a compact hardware solution Fig. 1. The compact design is realized by folding an open horn or cone structure inside a cylindrical or cubic formed pot. The noise enters the absorber via a permeable sheet at the large orifice on top of the pot. At the bottom of the pot a somewhat smaller orifice enables the acoustic coupling of both, pot and cone volume.

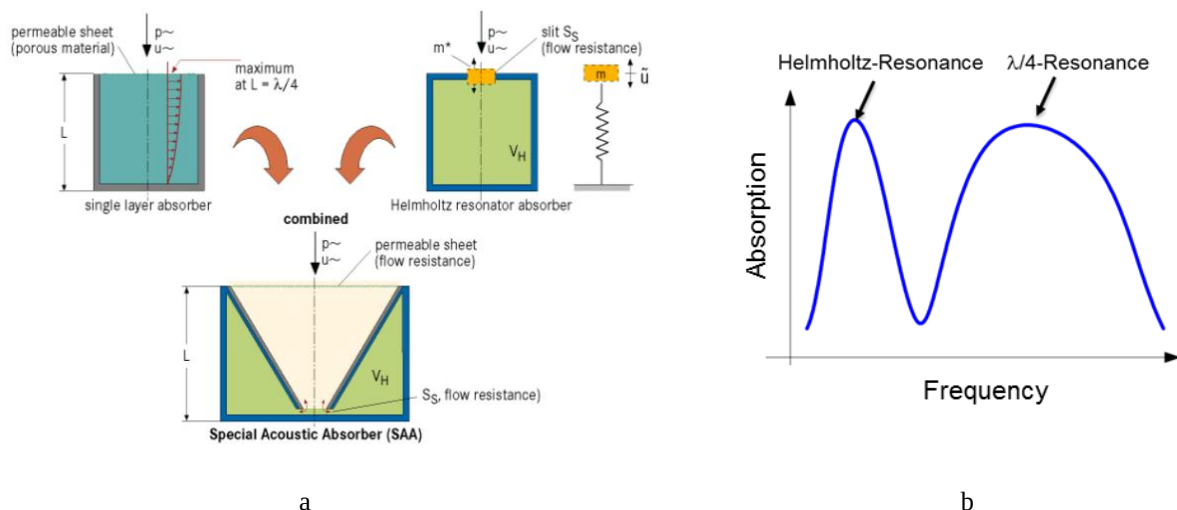


Fig. 1: Principal sketch of the special acoustic absorber (SAA) mechanism, a with typical absorption characteristic, b

In order to adjust the acoustic parameter as resonance frequencies and impedance to the specifications several approaches are possible. In the simplest case a lumped model for the Helmholtz part can be coupled with a 1D sound propagation model for the cone. Several assumptions had to be done on the acoustic properties as dimension of the coupling orifice and impedance of this orifice and permeable sheet. However, due to its simplicity such a validated model is suitable within an optimization process for the geometric dimensions. More complex analytical models¹ allow more accurate calculations of the resonance frequencies but are often restricted to special absorber designs as for instance cylindrical shape of the cone. In case of very complex structures numerical methods are preferable for final design optimisations. In that case ACTRAN-TM was used by the authors for the work presented in this paper. Depending on the actual SAA liner design concept the available liner volume may not be used completely from the SAA elements. The remaining space can additionally be used as a usual SDOF component. Due to the equal depth of this two absorber types (SAA and SDOF- $\lambda/4$), the resonance frequency of the SDOF- $\lambda/4$ part often becomes similar to the anti-resonance of the SAA. As a consequence of this combination a broad absorption bandwidth can be achieved, Fig. 2.

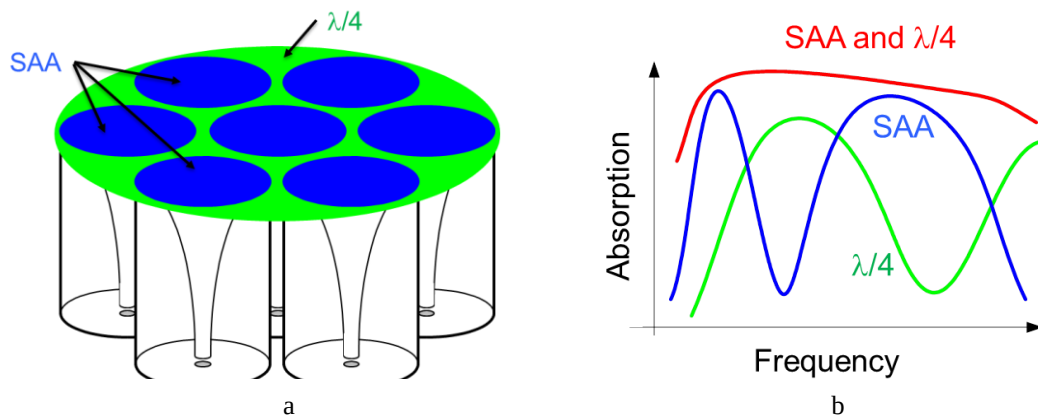


Fig. 2: Principal sketch of a SAA- $\lambda/4$ combination, a with typical absorption characteristic, b

III. SAA liner applications with different damping objectives

In the following SAA-based liner concepts for three different damping objectives with varying application conditions will be presented.

A. Typical 3DOF liner application

In order to reduce the emitted noise at an engine exhaust further a hot stream liner application was designed for a given engine. The liner specification was such that combustion noise with frequencies well below 800 Hz and turbine noise with frequencies well above 4 kHz had to be damped simultaneously, which spans the frequency range for the two components about one decade. The liner had to be included into the exhaust cone. Due to the low frequency of the combustion noise component and the limited space within the cone it was decided to make use of the SAA principle. With the aid of a classical SAA design a lower resonance frequency for the Helmholtz part can be realized within the range of some hundred Hz.

The design of the liner has been done in a usual way: From the specifications for the frequency ranges, which define the main spectral ranges with maximum absorption, its geometric dimension had been calculated. Based on this pre-design samples for impedance tube tests as well as for flow channel tests has been manufactured.

The build-up of the SAA- and $\lambda/4$ -elements is shown in Fig. 3, where for the flat sample each single part of the liner is visible. The flat sample was built with eight single absorber elements arranged within a building block offering an area to the flow channel of 200x100mm². Each single SAA element owing a circular cross sectional area is embedded into a single cuboid with rectangular cross section. This cuboid acts as an individual $\lambda/4$ -absorber with a resonance frequency according to its depth.

The flow channel arrangement is given in Fig. 4. This figure shows also the arrangement of the two in-situ microphones for direct extraction of the sample impedance as well as the microphone arrays in front and aft of the sample for modal decomposition and following calculation of absorption and transmission loss. Each of the two microphone arrays measures the sound field at the three axial sections. At each section four microphones are positioned on the larger lateral side of the rectangular flow tube and three microphones on the smaller horizontal side. Each array consists therefore of 21 microphones. The cross sectional area of the flow channel was 70x90mm². The open area ratio of the eight SAA-absorber elements relative to the lined surface was 28%.

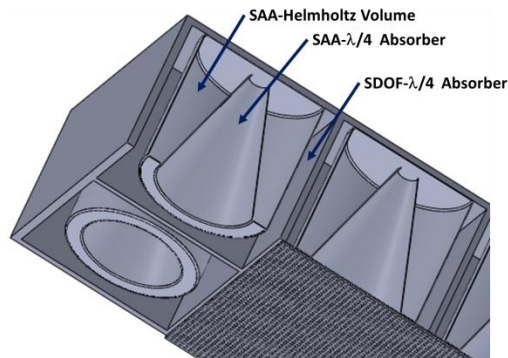


Fig. 3: Liner design with SAA- and SDOF- $\lambda/4$ elements for flow channel tests

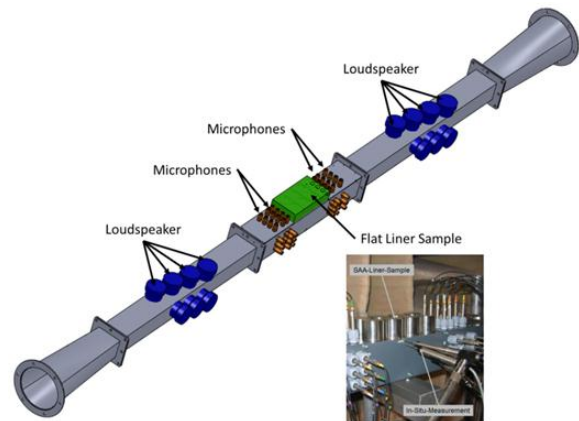


Fig. 4: Flow channel test set-up with loudspeaker and microphone arrangement and sample integration

Due to the low frequency tuning of the Helmholtz part the resulting orifice at the inner end of the cone had to be quite small compared to the outer orifice. Therefore the tapering of the SAA cone was quite high, which transforms the linear resistance part of the Helmholtz absorber to values which have to be accounted for during the design process. The liner should work at the engine exhaust. Therefore hot gas conditions according to a temperature of 500°C were specified. Due to this high temperature application it was decided to realize the necessary resistance by a pure perforate. Preliminary impedance estimations for different perforates had been done including mean flow and sound pressure level effects according to the literature². The impedance behaviour of this resistance sheet was measured at different sound pressure levels within both, an impedance tube and a DC-flow meter for the flow

resistivity as well as at different grazing flow Mach numbers up to $M = 0.3$ at ambient conditions. Typical results for measured resistance values are plotted in the diagram of Fig. 5 for the sound pressure level dependency and in the diagram of Fig. 6 for the grazing flow Mach number dependency. In case of varying Mach number the exciting sound pressure level was kept to a maximum in order to assure a sufficient signal to noise ratio.

Fig. 5 presents two curves, the green one represent the resistance values as measured and indicates a value for the linear part of $\Theta_{lin}/\rho c = 0.29$, resulting from the viscous losses at the small orifice at the inner part of the cone. With increasing sound pressure additional, pressure sensitive losses appear and dominate the resistance values. The blue curve represents the sound pressure dependant values only.

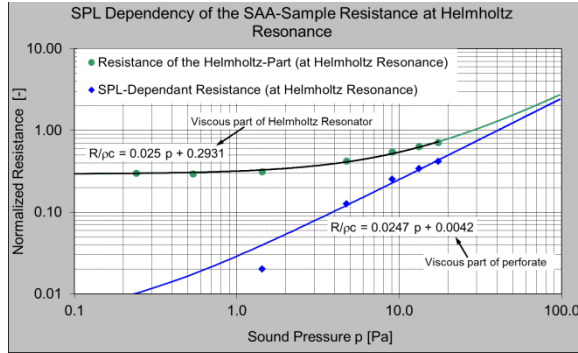


Fig. 5: Measured SAA resistance at the Helmholtz resonance vs. sound pressure

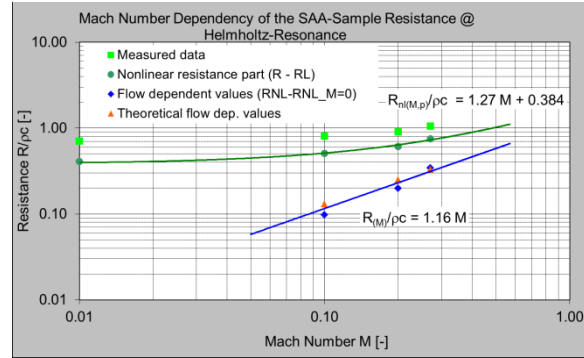


Fig. 6: Measured SAA resistance at the Helmholtz resonance vs. grazing flow Mach number

Pure grazing flow Mach number dependency, as given by the blue curve in the diagram of Fig. 6, was extracted by elimination of linear and non-linear, pressure sensitive resistance parts at zero flow from the measurement values. The remaining flow speed sensitive values are linear proportional to the flow Mach number and compares very well with results calculated according to Rice³:

$$\frac{\Theta_{(M)}}{\rho c} = \frac{0.3 M}{\sigma}$$

From these measurements the following empirical relation can be deduced for the pressure and grazing flow Mach number dependency of the Helmholtz resistance:

$$\frac{\Theta_{HR(p,M)}}{\rho c} = 0.29 + 0.025 p + 1.16 M$$

For the specified sound pressure level of about 130 dB and a grazing flow Mach number of about 0.3 a resistance of about 2.2 can be expected in case of ambient conditions. At hot gas conditions a somewhat higher value is expected as the viscous part will increase while the turbulence losses, responsible for the pressure and flow dependant part are expected not to be influenced remarkably. However, this measured linear proportionality between sound pressure and resistance does not agree with the pressure dependency as given by Motsinger and Kraft² where they found in case of perforated face sheets:

$$\frac{\Theta_{(p)}}{\rho c} \propto \sqrt{p}$$

A possible explanation for the measured pressure-resistance relation may be, that in case of the Helmholtz part resistance is located at two positions. First the sound wave must cross the perforate at the outer opening of the cone where one part of resistance is created. The second part is located at the small coupling orifice between the cone and the backing volume of the SAA. This assumption has to be tested within a further study.

Fig. 7 present the measured sound transmission loss within the flow channel for a flow Mach number of $M=0.2$. This transmission loss is represented twice. The blue curve gives the relation between transmitted and incident wave recalculated via a modal decomposition of the sound field at the two microphone array as shown in Fig. 4. The result

represented by the grey shaded region represent simply the ratio of the two sound pressure fields averaged over the microphones in front of and behind the sample. Both methods deliver very similar results. However, above a frequency of about 3 kHz the modal transmission loss gets noisy. In this diagram reduced sound transmission can be observed at the three working frequencies; the SAA-Helmholtz part, the SDOF $\lambda/4$ -part and the SAA- $\lambda/4$ -part. Additional transmission losses are visible in the vicinity of the cut-on frequencies for higher modes.

A typical measurement result for the absorbed sound pressure level within the flow tube is given in the diagram of Fig. 8. In this diagram two resonance ranges are visible, one at about 440 Hz which results from the Helmholtz part of the SAA and the other at about 1600 Hz resulting from the $\lambda/4$ -resonance of the SAA surrounding $\lambda/4$ -SDOF cube. This diagram shows that the absorption of the Helmholtz part diminishes continuously with increasing flow speed indicating that the resistance there is already above the optimum value for the no flow case. The quarter wave absorption increases from $M=0$ to $M=0.1$ and decreases for higher flow velocities.

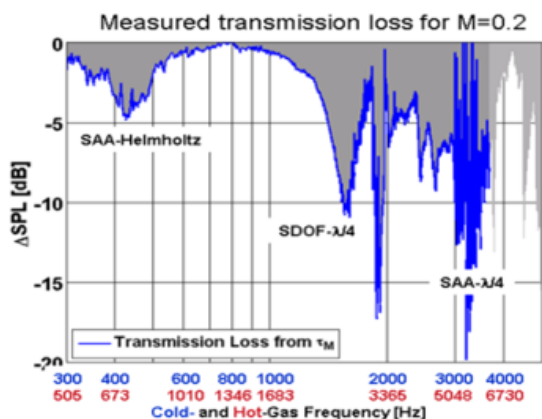


Fig. 7: Measured transmission loss of the flat sample within the flow channel at $M=0.2$

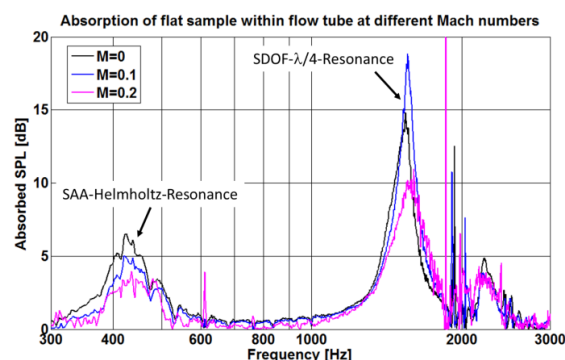


Fig. 8: Measured absorbed sound pressure level of the flat sample within the flow channel at varying grazing flow Mach number

The integration of the SAA elements into a real exhaust cone proved to be difficult. Based on the actual given cone geometry a design for integration was suggested as shown in Fig. 9a. This proposal however was not possible. Weight and stiffness constraints yielded to the solution as given in Fig. 9b. This implementation differs in two main points. First, the number of integrated SAA pots is remarkably less than proposed leading to a lower absorption area. Second, the remaining volumes around the cylindrical pots are quite large, especially is there lateral dimension greater than their depth. It must therefore be expected that they do no longer act as pure classical quarter wave absorbers. The complete lined cone is shown in Fig. 9c. All liner elements are made with heat-resistant metal.

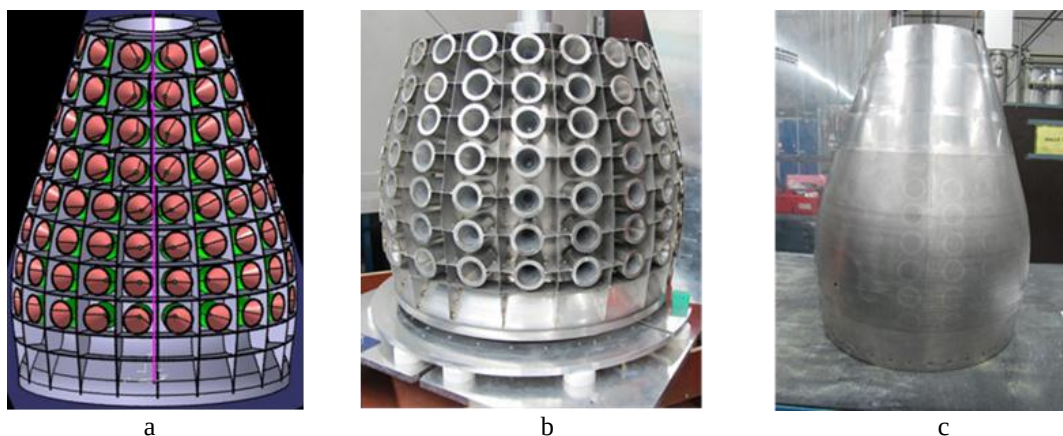


Fig. 9: SAA-Liner integration on an engine exhaust cone (Source: PFW Aerospace AG) (a: suggested design; b: lined part of the cone as manufactured; c: complete lined cone)

The drawings for the proposed liner integration as well as the whole liner manufacturing were done by PFW Aerospace AG. The lined cone has been tested by Rolls-Royce Deutschland GmbH on a real engine.

The acoustic test results did not fulfil the expectations. Three major reasons are identified for those results.

- The perforate chosen did not deliver enough resistance during the engine tests.
- The acoustic effective open area ratio of the SAA elements was too low.
- Leakage between the liner elements enabled later flow due to the usual pressure gradients inside the exhaust flow.

The leakage assumptions have been proved after the tests by feeding hot steam through the perforate into the liner and observe its diffusion with an infrared camera.

In order to clarify especially the question how the resistance of a perforate behave at hot flow conditions further tests on an adequate test are recommended.

B. Design for bypass application

In contrary to the liner specification for the exhaust cone was the frequency range to be damped less small for the bypass application. However, compared to a usual SDOF application the absorption bandwidth should be enlarged to lower frequencies. Furthermore the liner should offer optimised impedance over a greater frequency range as possible with an SDOF liner. In case of a single SDOF liner the tuning results from a compromise between absorption efficiency and bandwidth. Therefore, due to its DDOF behaviour the SAA principle was chosen also for a bypass liner application. For this application the specification for the frequency range to be lined was such that the Helmholtz resonance of the SAA had to be at about 2 kHz, its quarter wave resonance should be designed at about 4.3 kHz. Due to integration constraints of the liner into the test rig at ANECOM the depth of the liner should not exceed 1". Based on these guidelines the geometry of the SAA liner has been designed as shown in Fig. 10.

The main difference to the former exhaust liner design is that now the draft angle of the SAA cone nearly vanishes. The orifice at the inner cone end is only marginal less than the outer opening. The resistance created at this inner cone position is now reduced compared to the exhaust SAA design which diminishes the effect on the overall SAA impedance. Within Fig. 11, where measured impedance values are shown for a single SAA pot as used for the bypass liner, it becomes visible that without facing sheet the resistance is negligible, ($\Theta/\rho c < 0.03$). The total SAA resistance is specified by the facing sheet only.

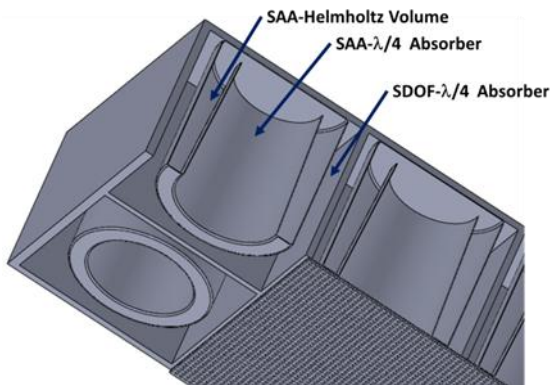


Fig. 10: SAA liner design for the bypass

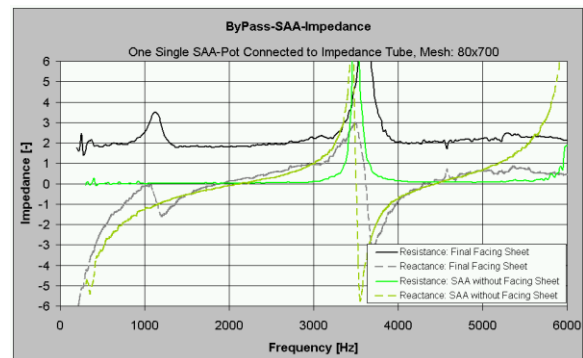


Fig. 11: Impedance tube measurement of the SAA-Pot with and without facing sheet

The facing sheet used for the bypass-liner was a combination of a wire mesh made of stainless steel and a thin perforate with 24% porosity made of carbon fibre which was oriented to the sound field. Wire mesh and perforate are glued. Within Fig. 11 a second anti-resonance becomes visible at about 1150 Hz. This additional and unwanted degree of freedom results as the SAA pot was made of plastic which enables the backing side of the cylindrical pot to vibrate. When the backing side is fixed this anti-resonance vanishes. A photograph of a single SAA pot is given in Fig. 12a, Fig. 12b shows the facing sheet.

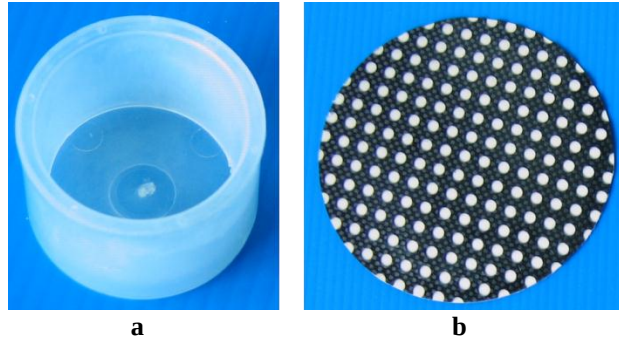


Fig. 12: SAA Pot, a and facing sheet, b of the bypass liner

The integration of the whole liner at the inner and outer section of the engine bypass on the ANECOM test rig is shown in Fig. 13a. A view from the backing side of the liner in direction to the facing sheet is given in Fig. 13b. This figure demonstrates the arrangement of the single SAA pots within separate cuboids.

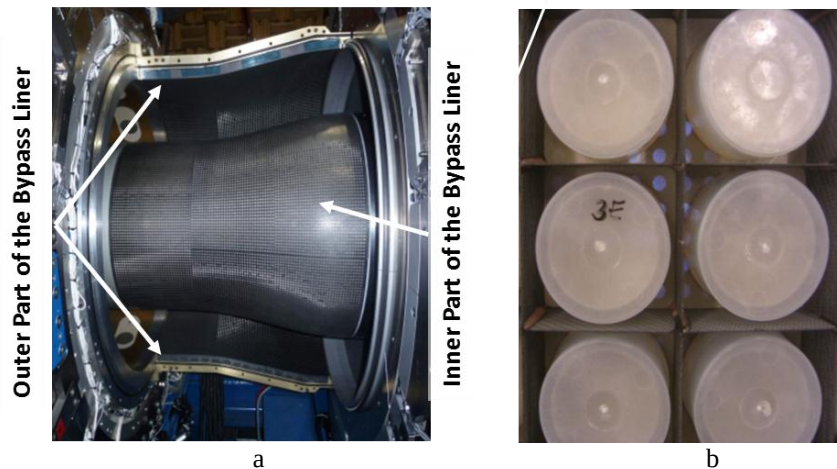


Fig. 13: Integrated liner into the bypass, a, and backing view into the liner, b

The manufacturing of this bypass liner was done by Wimmer Carbontechnik GmbH under the direction of PFW Aerospace AG. The measurements have been done at ANECOM in collaboration with Rolls-Royce Deutschland GmbH.

The measurement results achieved was very encouraging and may be presented in a future paper.

C. Volume optimized design for shrouded fan integration

In the two examples above the liner was designed such that the single SAA elements are surrounded by individual cuboids. In cases where only a limited frequency range had to be damped, the lower frequencies are quite small and where limited space is available a SAA liner concept without surrounded volumes may be advantageous. As a consequence of this the SAA elements cover the whole available liner area. The 3DOF behaviour reduces to a DDOF system however with increased number of SAA elements. The loss of one degree of freedom can be compensated for by choosing varying liner depths. Such a design was done for an application within a shrouded fan.

The frequency specification for the acoustic absorption interval started at 400 Hz and spread up to 2 kHz. The maximum liner depth should 60 mm not exceed. In order to achieve additional bandwidth the liner should be designed with slightly differing depths of the SAA elements. For flow tube tests of the liner design a flat sample was manufactured as shown in Fig. 14. The manufacturing was done by a laser sintering process in a rapid and comparatively cheap process. As facing sheet a 2-layer wire mesh was chosen with a resistance of about 1.7 pc.

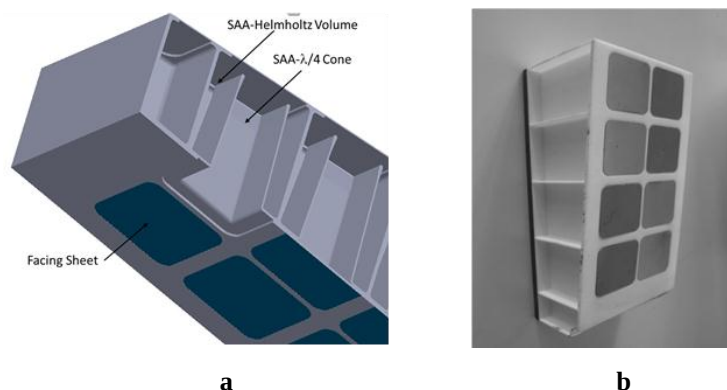


Fig. 14: SAA liner design for flat sample flow tube tests (drawing, a and manufactured sample, b)

The sample has been tested in the flow tube of Fig. 4. Due to the geometrical design of the liner its Helmholtz resonance frequency should be at about 500 Hz and its $\lambda/4$ -resonance at about 1500 Hz. These two specifications have been fulfilled as demonstrated in the measured transmission loss diagram of Fig. 15.

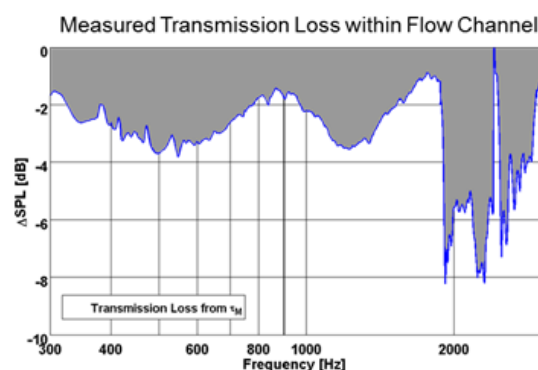


Fig. 15: Measured transmission loss in the flow tube of the liner sample

A possible integration of the upper design into the shroud is presented in Fig. 16 together with its inner set-up. The liner manufacturing is planned using a rapid prototyping process in order to minimize the necessary amount (time, cost, weight).

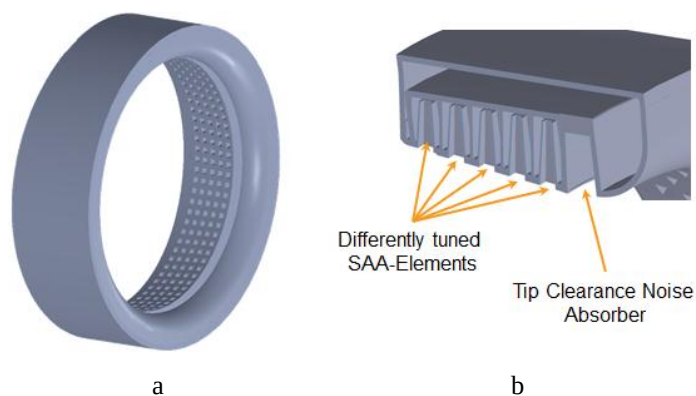


Fig. 16: SAA lined shroud, a and interior view, b

IV. Numerical simulations

All three application examples have been supported during the design phases by numerical simulations. These SAA liner simulations have been done using the commercial FEM tool ACTRAN/TM, working in frequency domain. The numerical models contains not only the geometrical properties of the test bench and liner samples which mainly determines the reactive acoustic liner properties but also necessary resistance information. Within the ACTRAN models this resistance information was realized by definition of adequate transfer admittances. An example of the 3D-mesh model for description of the single SAA element is presented in Fig. 17. For more accuracy in the description of the acoustic behaviour the SAA element was build up by three acoustic domains and also three areas for the implementation of the necessary transfer admittances for coupling of these domains. Furthermore due to the high model resolution the realistic orifice area ratio, as given in case of a perforate, can be accounted for.

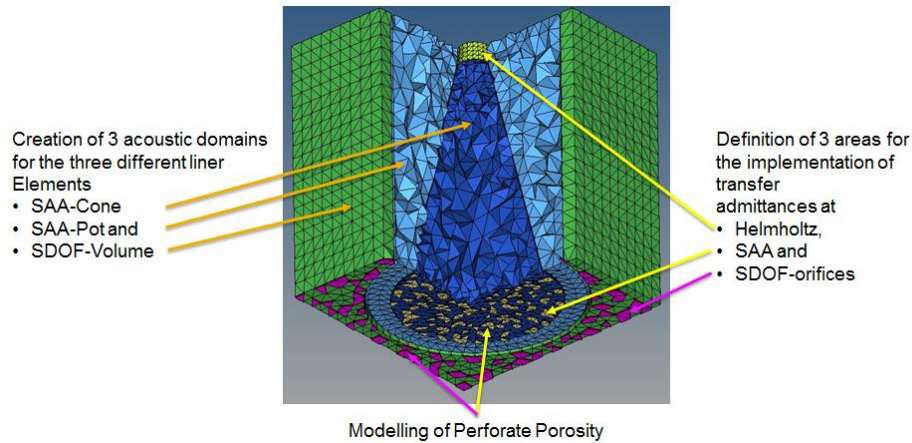


Fig. 17: Finite Element mesh of single SAA element for numerical simulations using ACTRAN

Using this numerical model for a single SAA-quarter wave element the impedance tube test set-up as well as the flow tube test set-up has been simulated. Both numerical model versions are given in Fig. 18 for the Kundt's tube simulation and Fig. 19 for the flow tube simulations. Besides the numerical study of geometric effects on the liner tuning and impedance it was also possible to extract information about the implementation of in-situ microphones for impedance extraction during the laboratory and engine tests. Using those test results the values of the transfer admittance used within the numerical models could be updated.

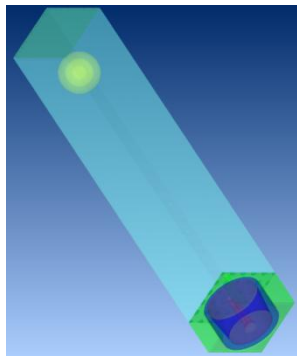


Fig. 18: 3D ACTRAN model of the isolated 3DOF SAA element with Kundt's tube apparatus and monopole source excitation

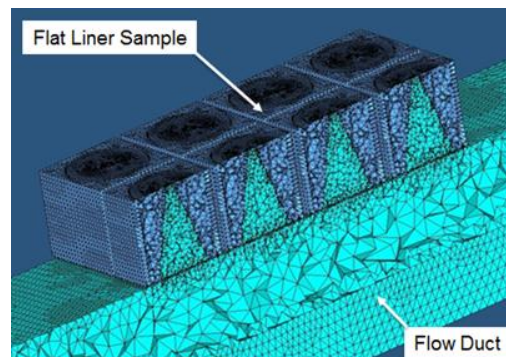


Fig. 19: Numerical model of the wind tunnel test configuration with 3DOF SAA-sample

The diagram of Fig. 20 compares those impedance data, (relative to p_c), as extracted with the help of an in-situ impedance measurement model for different situations. The data represent results extracted from pure acoustic excitation measurements as done after the liner integration into the bypass channel segments, results at two different engine settings (50% and 105%), simulation results and impedance data as measured inside the Kundt's tube (KT).

All these data converge sufficient well at the working frequency range of the liner. This result confirms also that the liner impedance does not depend on the exciting noise and flow fields as no significant difference can be seen between the engine settings as well as between engine tests and pure acoustic excited tests. The linear impedance behaviour of the wire mesh used as facing sheet may be the main reason for this independency of the liner from the excitation situation.

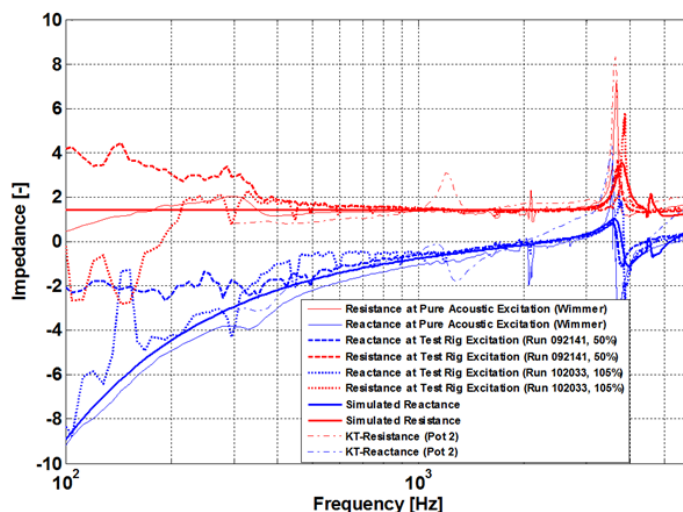


Fig. 20: Comparison between simulated and extracted impedance data for the bypass liner at different integration / excitation situations

The numerical model allows also insight into the spatial pressure distribution of the liner elements at any excitation frequency. Those representative sound pressure distributions are presented in Fig. 21 for the flow duct test set-up. For the three frequency ranges shown, the expected sound pressure distribution within the corresponding volume partitions are visible together with a remarkable noise reduction within the flow behind the liner segment, (sound propagates from left to right).

At the Helmholtz resonance frequency of 440 Hz strong pressure amplification inside the Helmholtz volume becomes visible relative to the exciting sound pressure whereas the pressure inside the cone compares to the exciting values with some phase shift only. At a frequency of 1600 Hz a resonance effect becomes visible only into the SAA surrounding cuboid. The pressure inside the cone reflects again that inside the flow tube and the Helmholtz volume already decouples. At the frequency of 2600 Hz the pressure inside the SAA cone is amplified relative to its excitation whereas the Helmholtz volume still decouples. This simulation confirms also that each of the three liner elements acts as expected.

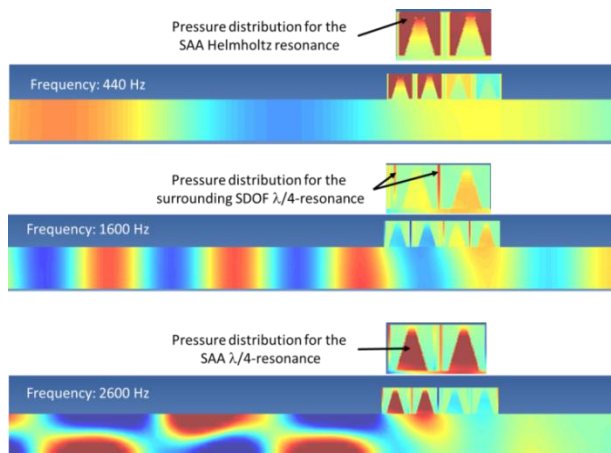


Fig. 21: Simulated sound pressure distribution within the sample volume partitions for their three degrees of freedom as simulated with the flow duct model

Based on those sample simulation and measurement results the acoustic effect of a volume optimized SAA liner integration into the shroud have been studied numerically. Fig. 22a shows the 3D numerical model for the lined shroud with integrated liner elements. As acoustic excitation a monopole source was positioned inside the fan plane near the shroud surface as shown in Fig. 22b.

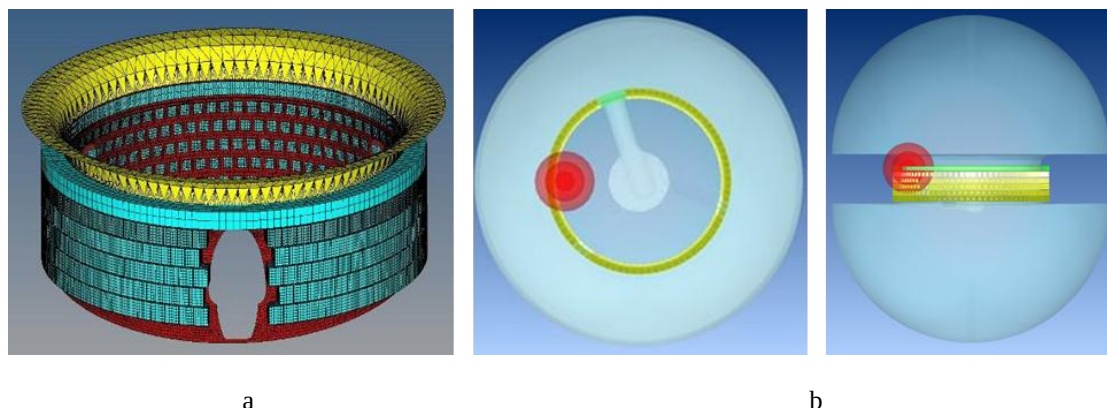


Fig. 22: 3D numerical model, a and aero-acoustic ACTRAN model, b of the lined shroud

The frequency of the source was swept within a defined range and the sound power emitted to the suction and pressure sides of the rotor have been integrated over the corresponding hemispheres located in the acoustic far-field. The numerical study has been done with two different resistance values for the facing sheet, 1.4 pc and 2 pc. The results are presented in Fig. 23 where the emitted sound power is plotted for the lined and unlined shroud. From these simulations broad noise reductions of 2-3 dB at the pressure side and 1-2 dB at the suction side can be expected. The differences for the two side's results from the fact that the rotor is located near the suction surface and the acoustically lined section of the shroud is located behind the rotor only. The results show also that the noise reduction is only weak dependant on the liner resistance chosen.

Further simulations with real aero-acoustic sources as extracted from the calculated unsteady flow field near the shroud surface are foreseen in the future.

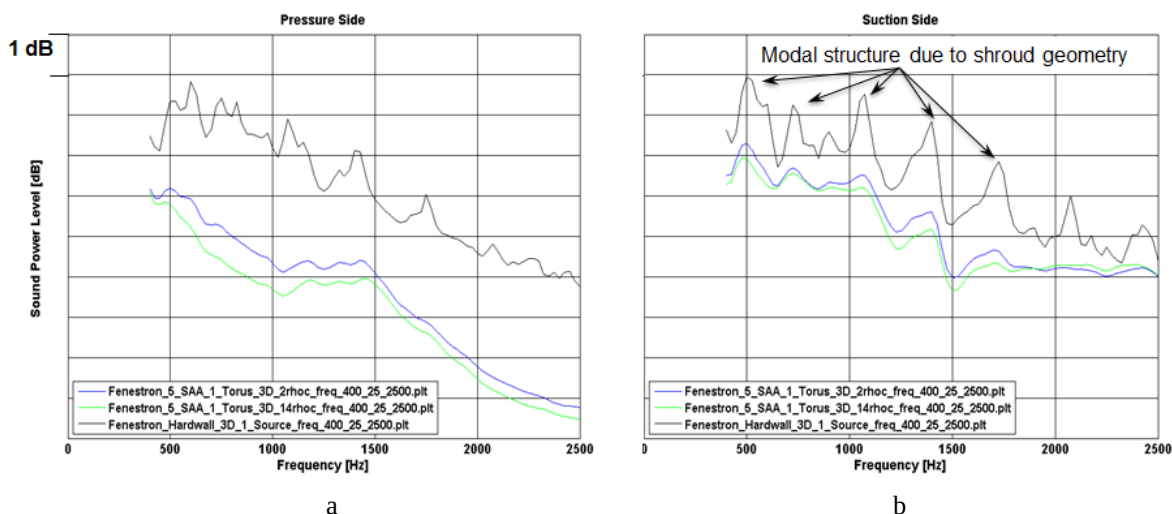


Fig. 23: Simulated noise reduction due to the SAA liner on pressure, a and suction, b side of the shroud

V. Conclusion

The aim of this paper was to present the liner integration activities on three different applications for the special acoustic absorber concept. Using the special acoustic absorber concept its possible application on three different design cases has been demonstrated. The three applications have been a hot stream liner for an engine exhaust, a liner within an engine bypass and a liner for the shroud of a fan. The SAA concept was chosen as it offer two main advantages: Due to its DDOF or even 3DOF concept it allows to damp a broad frequency range and due to its Helmholtz absorber element integrated it can be designed to low frequencies at limited space needed for integration. It could be shown that for each implementation an optimized design was possible, however weight and manufacturing amount can force to design modifications.

Flat sample measurements within impedance and flow tube confirmed the broadband absorption capability of this liner concept. However, measurements of the resistance at varying exciting sound pressure levels yielded to a linear proportionality which is in contrast to the literature. This question will be studied further.

During the design phase all three application examples have been supported by numerical simulations using the commercial tool ACTRAN-TM.

Acknowledgments

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